## 2.8 Classwork: Systems of Equations by Elimination — Markscheme

1. [Maximum mark: 9]

   *In each part: M1 for adding or subtracting the equations (after scaling, if used) so that one variable is eliminated, seen as the resulting one-variable equation. Substitution in place of elimination earns M0, then A1A1 for correct values. Correct values with no working (for example from the GDC) earn M0A1A1. FT the second value from the candidate's first value.*

   **(a)** $3x + 2y = 17,\ \ 5x - 2y = 7$

   Adding: $8x = 24$ **M1**

   $x = 3$ **A1**

   $3(3) + 2y = 17$, so $y = 4$ **A1**

   **(b)** $5x + 3y = 21,\ \ 4x - y = 27$

   $3\times$ second equation: $12x - 3y = 81$

   Adding: $17x = 102$ **M1**

   $x = 6$ **A1**

   $5(6) + 3y = 21$, so $3y = -9$ and $y = -3$ **A1**

   *Common error: scaling the left side only,* $12x - 3y = 27$*, gives* $17x = 48$*; M1 is not earned because the scaled equation is wrong. Accept eliminating* $x$ *first:* $4(5x + 3y) - 5(4x - y)$ *gives* $17y = -51$*.*

   **(c)** $3x + 4y = 14,\ \ 2x + 3y = 11$

   $2\times$ first: $6x + 8y = 28$; $\quad 3\times$ second: $6x + 9y = 33$

   Subtracting: $(6x + 9y) - (6x + 8y) = 33 - 28$ **M1**

   $y = 5$ **A1**

   $3x + 4(5) = 14$, so $3x = -6$ and $x = -2$ **A1**

   *Accept eliminating* $y$ *first:* $3(3x + 4y) - 4(2x + 3y)$ *gives* $x = 42 - 44 = -2$*. Common error:* $x = 2$ *(sign slip in* $14 - 20$*) earns the final A0.*

2. [Maximum mark: 12]

Survey points $(q,\ p)$: demand (16, 36) and (4, 72); supply (6, 33) and (20, 68). Quantity $q$ in hundreds of helmets per week, price $p$ in euros.

**(a)** $m = \dfrac{72-36}{4-16} = \dfrac{36}{-12} = -3$ **M1**

$36 = -3(16) + c$, so $c = 84$ **A1**

$p = 84 - 3q$ **AG**

*M1 for the gradient from the two demand points. A1 for $c = 84$ from a point. Only checking that the given line fits both points, with no gradient, earns M1A0. A gradient of $-\frac{1}{3}$ (run over rise) earns M0A0.*

**(b)** $m = \dfrac{68-33}{20-6} = \dfrac{35}{14}$ **M1**

$= 2.5$ **A1**

$33 = 2.5(6) + c$, so $c = 18$: $\quad p = 2.5q + 18$ **A1**

*Accept $p - 33 = 2.5(q - 6)$ or any equivalent form. FT the equation from the candidate's gradient. Wrong method: the gradient inverted, $\frac{14}{35} = 0.4$, earns M0A0, then A1 FT for $p = 0.4q + 30.6$.*

**(c)** $3q + p = 84$ and $2.5q - p = -18$ **A1**

Adding: $5.5q = 66$ **M1**

$q = 12$, so 1200 helmets per week **A1**

$p = 84 - 3(12) = 48$ (euros) **A1**

*A1 for both equations in the form $aq+bp = c$; accept $5q-2p = -36$ or any equivalent pair. M1 for adding or subtracting (after scaling, if used) so that one variable is eliminated. Accept eliminating $q$ first: $5(3q + p) - 3(5q - 2p)$ gives $11p = 528$. Substitution in place of elimination earns M0, then A1A1. Accept $q = 12$ (hundred) for the first value. FT from the candidate's supply line in (b).*

*Wrong methods: a sign slip, $2.5q - p = 18$, gives $5.5q = 102$, $q = 18.5$ (18.5454...), earning A0M1A0A0.*

**(d)** Substituting $p = 60$ into both equations **M1**

Demand: $60 = 84 - 3q$, so $q = 8$. Supply: $60 = 2.5q + 18$, so $q = 16.8$. **A1**

Surplus $= 16.8 - 8 = 8.8$ hundred, so 880 helmets per week **A1**

*The first A1 needs both quantities. Accept 8.8 (hundred) or 880 for the surplus. A negative answer, or calling it a shortage, earns the final A0. FT the surplus from the candidate's supply line in (b), provided supply exceeds demand. Wrong methods: comparing with the equilibrium quantity, $16.8 - 12 = 4.8$ or $12 - 8 = 4$, earns the final A0.*